QP CODE 25900002

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER
Discipline Specific Core Course (DSC) - MG2DSCAQC100 - ORNAMENTAL FISH CULTURE AND BREEDING
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the Blanks/ Name the Following
Answer all questions
Each question carries 1 marks

1. State one importance of Ornamental Fish Farming [K] / [CO2, CO5]

- **Economic Benefits**: High profitability in domestic and international markets.
- **Employment Generation**: Provides jobs in breeding, maintenance, and trade.
- **Export Potential**: Contributes to foreign exchange earnings.
- **Low Space Requirement**: Suitable for small-scale entrepreneurs.
- **Environmental Conservation**: Promotes captive breeding, reducing pressure on wild populations.
- **Aesthetic Value**: Enhances home and office decor through aquariums.
- **Educational Purposes**: Offers opportunities to learn about aquatic ecosystems.
- **Stress Relief**: Proven therapeutic benefits of maintaining aquariums.
- **Community Development**: Encourages cooperative ventures in rural areas.
- **Support for Allied Industries**: Boosts demand for equipment, feed, and accessories.

Mention any one benefit – 1 Mark

2. What is the art of designing underwater landscapes called? [U] / [CO2]

Aquascaping – 1 Mark

3. Write down the scientific name of pearl gourami. [K] / [CO4]

Trichogaster leerii or *Trichopodus leerii*

Both genus and species name – 1 mark

4. Give an example of submerged aquarium plant. [K] / [CO1]

Hornwort – *Ceratophyllum demersum*
Amazon Sword – *Echinodorus amazonicus*

Anacharis – *Egeria densa*
Java Fern – *Microsorum pteropus*
Dwarf Sagittaria – *Sagittaria subulata*
Vallisneria – *Vallisneria spiralis*
Cryptocoryne- *Cryptocoryne sp.*

Mention any one common name or scientific name or Genus name only – 1 Mark

5. What is the carbohydrate content percentage in herbivorous fish feed? [K] / [CO3]

25- 40 %

Any percentage within the above range – 1 Mark

6. Write the ideal pH range for most freshwater ornamental fishes. [K] / [CO1]

6.5 to 7.5 – 1 mark; Any other value within this range – 0.5 Marks

7. How often should water quality tests be conducted in aquariums? [K] / [CO2]

2-3 times per week; Weekly, Fortnightly or Monthly – 1 Mark for any of the periods mentioned

8. Write down the scientific name of a live bearer. [K] / [CO4]

Poecilia reticulata, *Poecilia sphenops*, *Xiphophorus maculatus* , *Xiphophorus helleri* etc.

Both Genus and Species names of any one – 1 Mark

9. What is a key nutritional requirement for fish broodstock? [K] / [CO4]

Protein , Essential Fatty acids ; Mention any one – 1 Mark

10. Name any two national funding agencies that support ornamental fish culture in India. [K] / [CO5]

1. PMMSY (Pradhan Mantri Matsya Sampada Yojana)
2. NABARD (National Bank for Agriculture and Rural Development
3. Fisheries and Aquaculture Development Fund: (FIDF) (Govt. of India)
4. MPEDA (The Marine Products Export Development Authority)
5. NATIONAL FISHERIES DEVELOPMENT BOARD (NFDB)

Mention any two funding agencies – 1 Mark

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

11. List the factors to consider when selecting an aquarium filter and explain why each factor is important. [K] / [CO1]

Factors to be considered while selecting an aquarium Filter

- Aquarium size or Tank Volume- Smaller filters may not be adequate , larger filters cause excess water flow

- Type of Filtration – Mechanical , Chemical and Biological Filtration- to remove particulate matter, dissolved gases and Ammonia respectively.

- Type of Filter – Hang on back,(Easy to install and maintain) Canister, (large tanks)Internal (compact and submerged)and Sponge Filters.(gentle water flow)

- Maintenance Requirement (easy to clean, maintain and assemble)

Any 3 factors and their explanation- 5 Marks

12. Give a detailed essay on design and construction of freshwater aquarium. [K] / [CO2]

Design

Aquarium Tank **–** Acrylic, All Glass, Framework aquarium
Sealants – Putty, Silicone sealant, Solvent cement
Size – Small, Medium and Large Tanks
Location – Avoid direct sunlight, level surface, Proximity to electrical outlets
Shape – Mostly Rectangular , allowing more surface area

Construction

- Filtration systems
- Equipments – Heating , Lighting
- Substrate- Gravel , Sand or pebbles

Setting

- Cleaning, Adding substrate and decorations, Filling with dechlorinated water, Installing Filter , Heater , Thermometer, Establishing bacterial colonies for filtration, Acclimatising fish and Maintenance.

Design – 2.5 Marks, Construction and Setting – 2.5 Marks; Total – 5 Marks

13. Explain the steps involved in the transportation of live ornamental fish. [K] / [CO4]

1.Preparation for Fish Transport

- Prophylactic treatment- remove infections, parasites and pathogens
- Starvation- prevent regurgitation of partially digested food and reduces excreta
-Prepackaging- Assessment of quality; weak and injured fishes removed.

2. Methods of Transportation

- Transporting without water in moist cloth or sack
- Open method – Earthen pots , Metal containers and Tanks
- Closed or Plastic bag method- HDPE, LDPE, Nylon, Polystyrene containers, filled with one part water and 3 parts pure oxygen, top of bag firmly twisted , folded and tied with thick thread or rubber bands.
- Packed in cardboard boxes, insulated with glasswool or cardboard pieces.

Preparation for Transport – 2 Marks; Methods of Transportation – 3 Marks

-14. How egg laying ornamental fishes are categorised? [K] / [CO4]

1. Fishes that lay eggs and allow them to hatch unattended.. Eg: Goldfish

a. **Egg Scatterers:** Fishes scatter eggs in water, while spawning. Adhesive eggs are attached to water plants and surfaces, while non adhesive eggs sink down
b. Egg Hangers: Eggs laid singly and attached to objects by a fine sticky thread.
c. Egg buriers: Eggs buried in soil
d. Egg storers: Eggs laid in mussel shells

2. Eggs are guarded by the parents , but the eggs are not in direct contact with the body surface. Eg: Clown fish, Angelfish.
a. Egg Anchorers: There are definite spawning sites on the surfaces of stones etc.
b.Egg splashers: Eggs laid on overhanging leaves and kept moist by splashing water
c. Egg Scoopers: Male scoops a depression in the substratum in which eggs are laid.
d. Egg Hiders: Eggs kept out of sight, between rocks or in holes scooped at the side of rocks.
e. Nest builders: Eggs protected in a nest formed of bubbles or twigs.

3. Eggs remain in direct contact with the body of the parent. Eg: *Tilapia*. They are of two types

a. Mouth Brooders: Eggs carried in the mouth of the parent
b. Egg Carriers: Eggs attached to some part of the body of the parent fish

Classification of 1st category – Eggs unattended – 2 Marks
Classification of 2nd category – Eggs guarded by parents – 2 Marks
Classification of 3rd Category – Eggs in contact with body of parents – 1 Mark
Total – 5 Marks

15. Write a short note on indigenous ornamental fishes of Kerala. [K] / [CO2]

Name	Characters	Habitat
Sahyadria denisoni (Miss Kerala, Denison Barb)	Known for its striking red stripe and active schooling behavior.	Rivers and streams in the Western Ghats.
Etroplus suratensis (Green Chromide, Pearl spot)	Greenish body with pearly spots, often kept in large aquariums.	Brackish and freshwater bodies.
Etroplus maculatus (Orange Chromide)	Bright orange body with small black spots.	Brackish and freshwater bodies.
Horabagrus brachysoma (Golden Catfish)	Golden body with dark markings.	slow-moving rivers, lakes, and reservoirs of Kerala.,muddy bottoms with vegetation
Parambassis ranga (Indian Glass Fish)	Transparent body allowing visibility of internal organs.	Streams and Estuaries
Trichogaster lalius (Dwarf Gourami)	Bright blue and red coloration with a peaceful temperament.	Slow-moving rivers and ponds.
Lepidocephalichthys thermalis (Travancore loach)	Small, bottom-dwelling fish with fascinating patterns.	Shallow streams and riverbeds.
Aplocheilus lineatus (Kerala killifish)	Shiny golden body with vibrant coloration.	Ponds and marshes.
Channa punctata (Spotted Snakehead)	Attractive spotted pattern, Air breathing	Slow-moving freshwater habitats.
Devario malabaricus (Malabar	beautiful fish with blue and gold	Fast-flowing streams and rivers.

Danio)	stripes.	
Rasbora daniconius (Blackline Rasbora)	Slim body with a prominent black lateral stripe.	Rivers, ponds, and lakes.
Puntius filamentosus (Blackspot barb) *Puntius fasciatus* (Melon Barb)	Vibrant colors and easy adaptability to aquariums.	
Mystus sp.	Small, attractive catfish with a sleek body and barbels.	Rivers, streams, Ponds , Lakes and Floodplain ecosystems
Garra mullya (Garra)	Known for their sucker-like mouths and algae-eating habits.	Fast flowing streams, Rivers and hill stream ecosystems.

Names and Notes on any five– 5 Marks , one mark each for complete details of one species

16. Explain the steps in induced breeding. [K] / [CO1]

1. Selection of Brooders:
Healthy and mature fish are chosen.
2. Hormonal Injection:
Hormones such as Ovaprim or HCG (Human Chorionic Gonadotropin) are injected to induce ovulation and spawning.
3. Spawning Setup:
Fish are placed in a specially prepared breeding tank with optimal water conditions.
4. Egg Collection and Fertilization:
Natural: Eggs are fertilized during spawning.
Artificial: Eggs and sperm are manually collected and mixed for fertilization.
5. Incubation and Hatching:
Fertilized eggs are incubated in safe conditions to ensure successful hatching.

Explanation of 5 steps – 5 Marks

17. Illustrate different categories of nutritional deficiency diseases in fishes with suitable examples

[K] / [CO2]

1. **Constipation**- occurs when dry feed is given to fishes
2. **Pin Heads:** The fry and fingerlings develop large heads and slender bodies, when exposed to frequent starvation.
3. **Lipoid Hepatic Degeneration**
- Usually caused by over feeding with feed containing rancid fats. , Liver becomes swollen and yellow brown in colour
- 4. **Rickets:**
- The disease is characterized by abnormal bone formation and caused by deficiency of Vitamin D, Rickets usually develop, when large quantities of coconut oil cake or maize is fed
5. **Soft shell syndrome:**
- (Exoskeleton or carpace) or External shells of prawns become weak, Caused by calcium and phosphorus deficiency in prawns.
6. **Vitaminosis A**
- Imbalance of Vitamin A is called Vitaminosis A
- Deficiency of Vitamin A or hypovitaminosis causes undergrowth, blindness, bleeding at the base of the fin and opaqueness of cornea.

Explanation of any 5 – 5 Marks

18. Describe various challenges faced by ornamental fish industry [K] / [CO5]

1. **Overexploitation of Wild Populations**:
   Excessive harvesting of ornamental fish from natural habitats can lead to biodiversity loss and ecosystem imbalance.
2. **Disease Management**:
   Frequent outbreaks of diseases in breeding and trading facilities can result in significant financial losses.
3. **Water Quality Issues**:
   Maintaining optimal water parameters is crucial but challenging, especially during transport and storage.
4. **High Mortality During Transport**:
   Stress, oxygen depletion, and poor packaging techniques often result in fish deaths during transit.
5. **Climate Change**:
   Rising temperatures and unpredictable weather patterns affect water sources and breeding cycles.
6. **Lack of Awareness and Training**:
   Many small-scale farmers and traders lack knowledge of proper breeding, handling, and marketing techniques.
7. **Regulatory Challenges**:
   Complex export/import regulations and quarantine requirements can delay shipments and increase costs.
8. **Competition from Imports**:
   Local breeders often struggle to compete with cheaper imported ornamental fish.
9. **Limited Market Access**:
   Difficulty in reaching international markets due to lack of infrastructure or promotional activities.
10. **Environmental Concerns**:
    Accidental release of non-native species into the wild can disrupt local ecosystems.
11. **Feed and Nutrition Costs**:
    High-quality feed is expensive, making it challenging for farmers to maintain profitability.
12. **Predation by Other Species**:
    Ornamental fish farms often face losses due to predation by birds or other animals.

    Explain any 5 points of the above or any related points ; 1 Mark for explanation of each point ; Total – 5 Marks

19. What is mechanical filtration? Explain its working principle, examples and benefits.[K] / [CO1]

Definition: a process that physically removes solid waste and particulate matter (like uneaten food, fish waste, and debris) from aquarium water by trapping it in a filter medium.

Working Principle: Aquarium water drawn through filter by pump or gravity and as it passes through the filter medium(sponge , pads or floss), solid particles are trapped and clear water returned to the Aquarium.

Examples: Container or Box Filter, Sponge Filter, Corner Filter containing filter media such as sponges, floss, sand, fibres, filter pads etc

Benefits: Improves water clarity, enhances biological filtration, prevents equipment damage due to clogging.

1.25 marks for each point; Total – 5 Marks

Part C
Essay
Answer any 1 questions
Each question carries 10 marks

20. Explain the international trade scenario of ornamental fish and India's role in the global market.[K] / [CO4] {Liberal Valuation recommended)

- The Business groups associated with Aquarium Fish trade include

Breeders- who rear or culture brood fishes
Growers - who buy small fry and rear them for 2-3 months till it is of a size to fetch good market price.
Carriers - who buy fry or fishes from breeders and/or growers and sell them to growers or retailers in the markets.
Retail shops or Hobby centers- They maintain link among the above and some who buy fishes for hobby
Ancillary/Supporting Unit- selling or trading related accessories required in maintenance of aquaria like aerators, air stones, filters, feeds, medicines, aquarium toys and aquatic plants etc.

Internal Fish Markets in India

- Turnover rates about 550 crore rupees
- Kolkata is the centre of Ornamental Fish Trade in India

Global Markets

- India ranks 31st among the exporting countries
- Singapore in the main trading hub of ornamental fishes in the world. Followed by Spain , Japan, Myanmar and Indonesia.
- USA is the major importing country in the world , followed by Germany, Japan, Netherlands and Singapore.
- Among the fish species angelfish and Goldfish are the most important. Top brands are the gold fish, guppy, molly, platy, cichlid, tetra, catfish, gourami, loach, and the barbs.
- The important species exported from India includes the Melon Barb (*Pethia conchonius)* and the Zebra danio (*Brachydanio rerio*).

Explanation of any 7 points from the above – 10 Marks

21. Write an essay on the different propagation methods of ornamental plants [K] / [CO1]

1. VEGETATIVE PROPAGATION- Propagation by using a part of the plant itself

- **Stem Cuttings**- using a cut portion of healthy stems
- **Runners or Stolons** – Using outgrowths called runners arising from leaf axils

- **Offsets**- Shorter and Stouter outgrowths growing close to parent plant
- **Rhizomes** – Underground stems, growing horizontally
- **Tubers** – Enlarged stem or roots that store nutrients
- **Bulb Propagation** – Specialised underground storage structure
- **Adventitious shoots** - plantlets that arise from any organ of the mother plant

2. SEXUAL PROPAGATION - involves the reproduction of plants from seeds, which are formed by the fusion of male and female gametes during fertilization.

3. MICROPROPAGATION or TISSUE CULTURE

- a method of propagating plants by growing cells, tissues or organs, in a sterile nutrient rich medium under controlled environmental conditions.

Explanation of Vegetative propagation with any 4 examples – 6 Marks
Explanation of Sexual Propagation – 2 Marks
Explanation of Micropropagation – 2 Marks
Total 10 Marks

22. Discuss the diversity and distribution of freshwater indigenous ornamental fishes in India.

[K] / [CO2]

1. **Melon Barb (Black Ruby Barb**) – *Puntius fasciatus* – western Ghats
2. **Western Ghats Loach (*Nemacheilus guentheri)*** – Western Ghats
3. **Denison's barb (*Sahyadria denisonii)*** – Western Ghats
4. **Blackspot barb (*Dawkinsia filamentosa*)-** Freshwater habitats across India
5. **Golden wonder Killifish- *Aplocheilus lineatus* –** Freshwater ponds and habitats all over India especially Western Ghats
6. **Rosy Barb (*Pethia conchonius)*** – Northern and Eastern parts of India; hill streams and associated floodplains
7. Indian glassfish (*Parambassis ranga*)- Northern and Eastern India, Ganges , Brahmaputra etc
8. **Dwarf Gourami (*Trichogaster lalius*)** – Norteastern India and West Bengal , Ganges, Brahmaputra etc.
9. **Indian Paradise Fish –*Macropodus cupanus*** – Southern India, hill streams and Freshwater ecosystems
10. **Spotted Snakehead (*Channa punctatus*)-** All over India in freshwater ecosystems
11. **Indian Leaf Fish (*Nandus nandus*)** - - Slow moving or still waters with vegetation- Ganga , Brahmaputra and Mahanadi river systems.
12. **Malabar Pufferfish (*Carinotetraodon travancoreus*)**- Endemic to western Ghats
13. **Striped Dwarf Catfish (*Mystus vittatus***) – Throughout India, slow moving waters with muddy substratum

Explain diversity and distribution of any 8 fishes – 10 Marks, any 6 fishes – 8 Marks, any 4- 6 marks, any 2- 4 Marks, any 1- 2 Marks